

Distribuce ve více dimenzích

- tenzorová testovací pole

- kompaktnost: Jaké uslovení?

Pohled video: jak souvisejí?

$\Rightarrow$ testovací oblasty s kompaktním nosičem

• $\mathcal{D}_k^l \dots$ testovací tenzorová pole typu (l, k)
s kompaktním nosičem

$$\Phi \in \mathcal{D}_k^l, \quad \Phi|_{\text{bod}}: \underbrace{\mathbb{V} \times \dots \times \mathbb{V}}_k \times \underbrace{\mathbb{V}^* \times \dots \times \mathbb{V}^*}_l \rightarrow \mathbb{R}$$

• $\mathcal{D}_l^{1,k}$... tenzorová distribuce typu (k, l)

$$\mathbf{T} \in \mathcal{D}_l^{1,k} \quad \Phi \in \mathcal{D}_k^l : \mathbf{T} : \Phi \rightarrow \mathbb{R}$$

$$\mathbf{T}(\Phi) = \langle \mathbf{T}; \Phi \rangle = \langle \mathbf{T}_{n_1 \dots n_l}^{m_1 \dots m_k}, \Phi_{n_1 \dots n_l}^{m_1 \dots m_k} \rangle$$

• regulární tenzorová distribuce

$$\langle \mathbf{T}_f; \Phi \rangle = \int_V f_{n_1 \dots n_l}^{m_1 \dots m_k} \Phi_{n_1 \dots n_l}^{m_1 \dots m_k} dV$$

Př. 1 Divergenční distribuce

$$\delta_{x_0} \in \mathcal{D}_0^0, \quad \Phi \in \mathcal{D}^0 : \delta_{x_0}(\Phi) = \Phi(x_0)$$

Př. 2

$$\chi_\Omega = \begin{cases} 1 & x \in \Omega \\ 0 & x \notin \Omega \end{cases} \quad \langle \chi_\Omega; \Phi \rangle = \int_\Omega \Phi dV$$

• gradient tenzorová distribuce

$$\text{motivace: } \langle \nabla \mathbf{T}_f; \Phi \rangle = \int_V (\nabla_a T_{n_1 \dots n_l}^{m_1 \dots m_k}) \Phi_{n_1 \dots n_l}^{m_1 \dots m_k} dV$$

$$\mathbf{T}_f \in \mathcal{D}_l^{1,k} \quad \Phi \in \mathcal{D}_k^{l+1}$$

$$= \int_V \nabla_a (T_{n_1 \dots n_l}^{m_1 \dots m_k} \Phi_{n_1 \dots n_l}^{m_1 \dots m_k}) dV = \int_V T_{n_1 \dots n_l}^{m_1 \dots m_k} \nabla_a \Phi_{n_1 \dots n_l}^{m_1 \dots m_k} dV$$

$$\int_{\partial V} (\mathbf{T} \cdot \Phi \dots) = 0 \quad - \langle \mathbf{T}; \nabla \cdot \Phi \rangle$$

$$\text{DEF: } \langle \nabla \mathbf{T}; \Phi \rangle = - \langle \mathbf{T}; \nabla \cdot \Phi \rangle$$

Př. • $\mathbb{E}^3$

• gradient: $\mathbf{T} \in \mathcal{D}_0^0, \quad \vec{\Phi} \in \mathcal{D}_1^0$

$$\langle \nabla \mathbf{T}; \vec{\Phi} \rangle = - \langle \mathbf{T}; \nabla \cdot \vec{\Phi} \rangle$$

• divergence $\langle \nabla \cdot \vec{T}; \Phi \rangle = - \langle \vec{T}; \nabla \Phi \rangle$

$$\nabla \cdot (\Phi \mathbf{T}) = \mathbf{T} \cdot \nabla \Phi + \Phi \nabla \cdot \mathbf{T}$$

• rotace: $\langle \nabla \times \vec{T}; \vec{\Phi} \rangle$

$$\nabla \cdot (\vec{T} \times \vec{\Phi}) = \vec{T} \cdot \vec{\Phi} \times \overset{*}{\nabla}$$

$$= - \vec{T} \cdot (\nabla \times \vec{\Phi}) + (\vec{\Phi} \times \nabla) \cdot \vec{T}$$

$$= - \vec{T} \cdot (\nabla \times \vec{\Phi}) + \vec{\Phi} \cdot (\nabla \times \vec{T})$$

$$\Rightarrow \langle \nabla \vec{T}; \vec{\Phi} \rangle = \langle \vec{T}; \nabla \times \vec{\Phi} \rangle$$

Plošné distribuce v $\mathbb{E}^3$

Shalduní

$$\delta_\Sigma \quad \langle \delta_\Sigma; \Phi \rangle = \int_\Sigma \Phi dS$$

$$= \int_{\mathbb{E}^3} \Phi \delta_\Sigma dV$$

Vektorová

$$\vec{\delta}_\Sigma \quad \langle \vec{\delta}_\Sigma; \vec{\Phi} \rangle = \int_\Sigma \vec{\Phi} \cdot d\vec{S}$$

$$\vec{n} \dots \text{normální } \Sigma \quad d\vec{S} = \vec{n} dS$$

$$\Rightarrow \langle \vec{\delta}_\Sigma; \vec{n} \cdot \vec{\Phi} \rangle = \langle \vec{\delta}_\Sigma; \vec{\Phi} \rangle$$

$$\Rightarrow \vec{\delta}_\Sigma = \delta_\Sigma \vec{n} \quad \wedge \quad \delta_\Sigma = \vec{n} \cdot \vec{\delta}_\Sigma$$

Křivkové distribuce v $\mathbb{E}^3$

Shalduní

$$\gamma \dots \text{křivka} \quad \gamma: (0,1) \rightarrow \mathbb{E}^3$$

$$\langle \delta_\gamma; \Phi \rangle = \int_\gamma \Phi ds$$

Vektorová

$$\langle \vec{\delta}_\gamma; \vec{\Phi} \rangle = \int_\gamma \vec{\Phi} \cdot d\vec{s} = \int_\gamma \vec{\Phi} \cdot \vec{t} d\vec{s} = \langle \delta_\gamma; \vec{t} \cdot \vec{\Phi} \rangle$$

$\vec{t} \dots$ tečnový vektorový pole

$$\|\vec{t}\| = 1$$

$$\Rightarrow \vec{\delta}_\gamma \stackrel{!}{=} \vec{t} \delta_\gamma$$

$$\delta_\gamma \stackrel{!}{=} \vec{t} \cdot \vec{\delta}_\gamma$$

Vztah k integrálnímu větě

$\Omega \dots$ oblast v $\mathbb{R}^3$

$$1) \langle \vec{\nabla} \chi_\Omega; \vec{\Phi} \rangle = - \langle \chi_\Omega; \vec{\nabla} \cdot \vec{\Phi} \rangle = - \int_\Omega \vec{\nabla} \cdot \vec{\Phi} dV \stackrel{\text{Gaus}}{=} - \int_{\partial\Omega} \vec{\Phi} \cdot d\vec{S} = - \langle \vec{\delta}_{\partial\Omega}; \vec{\Phi} \rangle$$

$$\Rightarrow \vec{\nabla} \chi_\Omega = - \vec{\delta}_{\partial\Omega}$$

$$2) \langle \vec{\nabla} \times \vec{\delta}_\Sigma; \vec{\Phi} \rangle = \langle \vec{\delta}_\Sigma; \vec{\nabla} \times \vec{\Phi} \rangle = \int_\Sigma (\vec{\nabla} \times \vec{\Phi}) \cdot d\vec{S}$$

$$\stackrel{\text{Stokes}}{=} \int_{\partial\Sigma} \vec{\Phi} \cdot d\vec{s} = \langle \vec{\delta}_{\partial\Sigma}; \Phi \rangle$$

$$\Rightarrow \vec{\nabla} \times \vec{\delta}_\Sigma \stackrel{!}{=} \vec{\delta}_{\partial\Sigma}$$

$$3) \langle \vec{\nabla} \cdot \vec{\delta}_\Sigma; \Phi \rangle = - \langle \vec{\delta}_\Sigma; \nabla \Phi \rangle = - \int_\Sigma \nabla \Phi \cdot d\vec{S}$$

$$= - \int_\Sigma \vec{t} \cdot \vec{\nabla} \Phi dS = - \int_\Sigma \frac{d\Phi}{ds} ds$$

$$= - [\Phi(\gamma(1)) - \Phi(\gamma(0))] = - \langle \delta_{\gamma(1)} - \delta_{\gamma(0)}; \Phi \rangle$$

$$\Downarrow$$

$$\begin{matrix} \nabla & \nabla \times & \nabla \cdot \\ \downarrow & \downarrow & \downarrow \\ \partial & \partial & \partial \end{matrix}$$

$$\vec{\nabla} \times \vec{\nabla} \chi_\Omega = 0 \quad \Leftrightarrow \quad \partial \partial = 0$$

$$\text{funkce} \quad y = y(\vec{x}) : \begin{cases} y > 0 & x \in \Omega \\ y \leq 0 & x \notin \Omega \end{cases}$$

$$\Rightarrow \chi_\Omega = \theta(y(\vec{x}))$$

$$\vec{\nabla} \chi_\Omega = \vec{\nabla} \theta(y(\vec{x})) = \vec{\nabla} y(\vec{x}) \delta(y)$$

$$= - \vec{\delta}_{\partial\Omega}$$

$$\delta_{\partial\Omega} = \vec{n} \cdot \vec{\delta}_{\partial\Omega}$$

$$= \vec{n} \cdot \vec{\nabla} y \delta(y)$$

$$= - \frac{\partial y}{\partial n} \delta(y)$$

$$\vec{n}$$

$$\vec{n}_+$$

$$\vec{n}_-$$

$$\vec{n}$$

Formální spříslušnost δ -distribucí

$$\delta_\gamma = \int_{\gamma \in \gamma} \delta_z dz \quad \vec{\delta}_\gamma = \int_{\gamma \in \gamma} \delta_z d\vec{s}$$

$$\delta_\Sigma = \int_{\Sigma \in \Sigma} \delta_z dS \quad \vec{\delta}_\Sigma = \int_{\Sigma \in \Sigma} \delta_z d\vec{S}$$

$$\langle \int_{\Sigma \in \Sigma} \delta_z d\vec{S}; \vec{\Phi} \rangle = \int_{\Sigma \in \Sigma} \langle \delta_z \vec{n}; \vec{\Phi} \rangle dS =$$

$$= \int_{\Sigma \in \Sigma} \langle \delta_z \vec{n} \cdot \vec{\Phi} \rangle dS$$

$$= \int_{\Sigma \in \Sigma} \vec{n} \cdot \vec{\Phi}(z) dS = \int_\Sigma \vec{\Phi} \cdot d\vec{S} = \langle \vec{\delta}_\Sigma; \Phi \rangle$$

$$\Omega^-$$

$$\Omega^+$$

$$\Sigma$$

$$\vec{n}_+$$

$$\vec{n}_-$$

$$\vec{n}$$

$$f = \chi_+ f_+ + \chi_- f_-$$

$$\vec{F} = \chi_+ \vec{F}_+ + \chi_- \vec{F}_-$$

$$[f] = (f_+ - f_-)|_\Sigma$$

$$[\vec{F}] = (\vec{F}_+ - \vec{F}_-)|_\Sigma$$

$$1) \nabla f = \chi_+ \nabla f_+ + \chi_- \nabla f_- - f_+ \delta_\Sigma + f_- \delta_\Sigma$$

$$= \dots + f_+ \vec{\delta}_\Sigma - f_- \vec{\delta}_\Sigma$$

$$= \dots + [f] \vec{\delta}_\Sigma$$

$$2) \vec{\nabla} \cdot \vec{F} = \chi_+ \vec{\nabla} \cdot \vec{F}_+ + \chi_- \vec{\nabla} \cdot \vec{F}_- - \vec{\delta}_{\partial\Omega^+} \cdot \vec{F}_+ - \vec{\delta}_{\partial\Omega^-} \cdot \vec{F}_-$$

$$= \dots + [\vec{F}] \cdot \vec{\delta}_\Sigma = \dots + [\vec{F}] \cdot \vec{n} \delta_\Sigma$$

$$3) \vec{\nabla} \times \vec{F} = \chi_+ \vec{\nabla} \times \vec{F}_+ + \chi_- \vec{\nabla} \times \vec{F}_- - \vec{\delta}_{\partial\Omega^+} \times \vec{F}_+ - \vec{\delta}_{\partial\Omega^-} \times \vec{F}_-$$

$$= (\vec{\nabla} \times \vec{F})_{\text{reg}} + \vec{\delta}_{\partial\Omega} \times [\vec{F}]$$

$$S = \sigma \delta_\Sigma$$

$\uparrow$ usmírná dipólová hustota distribuce

$$S_{\text{reg}} \in L^1_{loc}$$

$$Q = \int_{\mathbb{R}^3} S_{\text{reg}} \mathbb{1} dV = \langle S_{\text{reg}}, \mathbb{1} \rangle$$

$$\Rightarrow \langle S, \mathbb{1} \rangle = \langle \sigma \delta_\Sigma, \mathbb{1} \rangle = \langle \delta_\Sigma, \sigma \mathbb{1} \rangle$$

$$= \int_\Sigma \sigma \mathbb{1} dS = Q$$

Dipól

$$q, \vec{d} \quad \vec{p} = q \vec{d} \Rightarrow \vec{d} = \frac{\vec{p}}{q}$$

$$\vec{p}$$

$$\vec{d}$$

$$\vec{p}$$

$$S_{\text{DIP}} = \lim_{q \rightarrow \infty} q (\delta_{\vec{r}_+ + \frac{\vec{p}}{q}} - \delta_{\vec{r}_- - \frac{\vec{p}}{q}})$$

$$= \lim_{q \rightarrow \infty} q (\delta_{\vec{r}_+ + \frac{\vec{p}}{2q}} - \delta_{\vec{r}_- - \frac{\vec{p}}{2q}})$$

$$= \lim_{q \rightarrow \infty} q \left(\phi(\vec{r}_+ + \frac{\vec{p}}{2q}) - \phi(\vec{r}_- - \frac{\vec{p}}{2q}) \right)$$

$$= \vec{p} \cdot \nabla \phi|_{\vec{r}} = - \langle \vec{p}; \nabla \delta_{\vec{r}}; \phi \rangle$$

Green v $\mathbb{R}^N$

$$\Delta_x G(x, x') = \underbrace{\delta(x - x')}_{\delta \vec{x}'}, \quad |\vec{x} - \vec{x}'| = r$$

$$1) \quad d=1 \quad G = \frac{1}{2} r$$

$$2) \quad d=2 \quad G = \frac{1}{2\pi} \ln r \Rightarrow \langle \Delta \frac{1}{2\pi} \ln r, \Phi \rangle$$

$$3) \quad d=3 \quad G = -\frac{1}{4\pi} \frac{1}{r}$$

$$4) \quad d \geq 3 \quad G = \frac{-1}{(d-2) S_d} \frac{1}{r^{d-2}} \quad S_d = \frac{2 \sqrt{\pi^d}}{\Gamma(\frac{d}{2})}$$

$$\Delta G = \delta_{\vec{x}'}, \quad |(-\vec{p} \cdot \vec{\nabla})$$

$$\Delta(-\vec{p} \cdot \nabla G) = -\vec{p} \cdot \nabla \delta_{\vec{x}'} = S_{\text{DIP}} \vec{x}'$$

$$\Phi_{\text{DIP}} = -\vec{p} \cdot \nabla G$$

$$= \vec{p} \cdot \frac{1}{4\pi r^3} = \frac{-1}{4\pi} \frac{\vec{p} \cdot \vec{r}}{r^3}$$